1. Electricity Theft Detection

🎯 Difficulty

Intermediate–Advanced

⏱️ Time to Complete

10–14 days

📊 Kaggle Dataset

Search on Kaggle for datasets similar to:

- **Electricity Load Diagrams Dataset**
- **Smart Meter Energy Consumption Dataset**
-

Example search terms:

```
electricity consumption smart meter dataset
electricity load diagrams kaggle
```

🧠 ML Techniques

- Anomaly detection
- Isolation Forest
- Random Forest
- Outlier detection

🔧 Features

- hourly electricity usage
- seasonal consumption
- day of week
- household type

📁 Project Components

1. Data preprocessing
2. Feature engineering
3. Anomaly detection model
4. Visualization of suspicious users
5. Feature importance analysis

⭐ Resume Value

Utilities and smart grid analytics are **very real-world industry problems**.

---

2. Crop Yield Prediction

🎯 Difficulty

Beginner–Intermediate

⏱️ Time

5–7 days

📊 Kaggle Dataset

Search for:

- **Crop Yield Prediction Dataset**
- **Global Crop Production Dataset**

Popular dataset search:

```
crop yield prediction kaggle
```

🧠 Algorithms

- Linear Regression
- Random Forest
- Gradient Boosting

🔧 Features

- rainfall
- temperature
- soil type
- fertilizer usage
- pesticide usage

📁 Final Project

- Data cleaning
- EDA visualizations
- Regression models comparison
- Feature importance

⭐ Resume Value

Shows **ML in agriculture analytics** 🌾 .

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3. Fake Product Review Detection

🎯 Difficulty

Intermediate

⏱️ Time

7–10 days

📊 Kaggle Dataset

Search:

- **Amazon Fake Reviews Dataset**
- **Fake Review Detection Dataset**

Example search:

```
fake product review detection kaggle
```

🧠 Algorithms

- Logistic Regression
- Random Forest
- Naive Bayes

🔧 Features

- review length
- sentiment polarity
- reviewer frequency
- rating pattern

📁 Final Project

1. Text preprocessing
2. Feature extraction (TF-IDF)
3. Classification models
4. Confusion matrix

⭐ Resume Value

Demonstrates **NLP + classification**.

---

4. Traffic Accident Severity Prediction

🎯 Difficulty

Intermediate

⏱️ Time

7–10 days

📊 Kaggle Dataset

Search:

`traffic accident severity dataset kaggle`

Popular dataset:

- US Accidents dataset

🧠 Algorithms

- Decision Tree
- Random Forest
- XGBoost

🔧 Features

- weather
- road condition
- time of accident
- vehicle type

📁 Final Project

- Accident distribution analysis
- Severity prediction model
- Feature importance visualization

⭐ Resume Value

Good example of **public safety analytics**.

---

5. Hospital Bed Demand Prediction

🎯 Difficulty

Advanced

⏱️ Time

12–15 days

📊 Kaggle Dataset

Search:

`hospital admissions dataset kaggle`

`hospital patient demand dataset`

Datasets often used:

- Hospital admissions datasets
- Healthcare demand datasets

🧠 Algorithms

- Random Forest Regression
- Gradient Boosting
- Time series models

🔧 Features

- patient admission count
- seasonal diseases
- historical hospital load

📁 Final Project

- time-series visualization
- demand forecasting
- prediction dashboard

⭐ Resume Value

Strong **healthcare analytics project**.

---

6. Student Career Path Prediction

🎯 Difficulty

Beginner–Intermediate

⏱️ Time

5–6 days

📊 Kaggle Dataset

Search:

```
student performance dataset kaggle
```

```
career prediction dataset
```

🧠 Algorithms

- Logistic Regression
- Random Forest
- Decision Tree

🔧 Features

- marks
- interests
- technical skills
- personality traits

📁 Final Project

- data visualization
- classification models
- recommendation output

⭐ Resume Value

Shows **education analytics**.

---

7. Food Waste Prediction in Restaurants

🎯 Difficulty

Intermediate

⏱️ Time

6–8 days

📊 Kaggle Dataset

Search:

```
food waste dataset kaggle
```

```
restaurant waste dataset
```

🧠 Algorithms

- Random Forest
- Gradient Boosting
- Linear Regression

🔧 Features

- customer count
- day of week
- menu items
- event days

🗂️ Final Project

- waste analysis
- regression prediction model
- waste reduction insights

⭐ Resume Value

Great **sustainability ML project** 🌍.

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8. Taxi Demand Prediction

🎯 Difficulty

Intermediate–Advanced

⏱️ Time

10–12 days

📊 Kaggle Dataset

Search:

`nyc taxi demand dataset kaggle`

`taxi trip dataset`

Popular dataset:

- NYC Taxi Trip dataset

🧠 Algorithms

- Random Forest
- Gradient Boosting
- Time series forecasting

🔧 Features

- time of day
- weather
- holidays
- location zone

📁 Final Project

- demand heatmap
- forecasting model
- demand visualization

⭐ Resume Value

Very strong **urban mobility analytics project**.

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9. Insurance Claim Fraud Detection

🎯 Difficulty

Intermediate

⏱️ Time

7–9 days

📊 Kaggle Dataset

Search:

`insurance fraud detectin dataset kaggle`

🧠 Algorithms

- Logistic Regression
- Random Forest
- Gradient Boosting

🔧 Features

- claim amount
- accident type
- customer history
- claim frequency

📁 Final Project

- fraud analysis
- classification models
- ROC curve evaluation

⭐ Resume Value

Shows **financial fraud detection skills**.

🔟 Dynamic Parking Price Prediction

🎯 Difficulty

Advanced

⏱️ Time

10–12 days

📊 Kaggle Dataset

Search:

```
parking demand dataset kaggle
```

```
parking occupancy dataset
```

🧠 Algorithms

- Linear Regression
- Random Forest
- Gradient Boosting

🔧 Features

- parking occupancy
- location
- time
- events nearby

📁 Final Project

- demand analysis
- pricing model
- price prediction system

⭐ Resume Value

Shows **smart city analytics**.

📊 Summary Table

Project	Difficulty	Time
Electricity Theft Detection	Advanced	10–14 days
Crop Yield Prediction	Beginner	5–7 days
Fake Review Detection	Intermediate	7–10 days
Traffic Accident Severity	Intermediate	7–10 days
Hospital Bed Prediction	Advanced	12–15 days
Student Career Prediction	Beginner	5–6 days
Food Waste Prediction	Intermediate	6–8 days
Taxi Demand Prediction	Intermediate–Advanced	10–12 days
Insurance Fraud Detection	Intermediate	7–9 days
Dynamic Parking Pricing	Advanced	10–12 days

⭐ Best 3 Projects for Your Portfolio

If you want **maximum impact**, build these:

1️⃣ Taxi Demand Prediction
2️⃣ Insurance Fraud Detection
3️⃣ Electricity Theft Detection

These show **time series + classification + anomaly detection**.